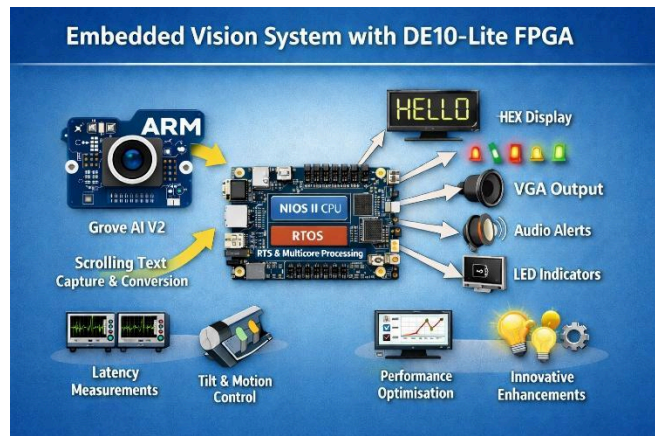


Real-Time Embedded Vision Using Arm – Grove AI V2 and NIOS based Multitasking System on FPGA

1. Description:

This mini-project involves the development of a real-time embedded vision system that integrates edge-AI processing with FPGA-based embedded control. The system combines a Grove AI V2 vision module with a Altera DE10-Lite FPGA board to demonstrate vision-enabled multitasking embedded system design. The Grove AI V2 module is an edge-AI vision device built around an ARM-based processor optimized for

on-device machine vision tasks. It can perform real-time visual inference, including text recognition, object detection, and image classification, without relying on external cloud computation. By executing vision algorithms directly on the module, the Grove AI V2 significantly reduces latency, bandwidth requirements, and system power consumption, making it suitable for embedded and edge-AI applications. Student groups (maximum of four students per group) should utilize the module to capture scrolling text displayed at varying speeds and convert the visual information into digital text data in the form of sentences or short paragraphs (up to 50 words). The extracted text data is then transmitted to the FPGA system using suitable NIOS peripheral interface.



The DE10-Lite FPGA board serves as the main embedded processing platform. Students are required to implement more than one NIOS II soft-core processor and design a multitasking or multi-controller architecture within a single FPGA fabric. In this project, multiple NIOS II processors are instantiated, with each processor responsible for a specific task, such as acquiring data from the Grove AI module via SPI, controlling visual and audio output peripherals, and managing system interaction and display logic. The processed information is presented using multiple FPGA-connected peripherals, including HEX displays, LEDs, VGA output, and an audio indicator such as a buzzer or speaker with programmable frequency output. The system may also respond to board orientation or motion inputs, obtained through the on-board accelerometer on the DE10-Lite, by dynamically modifying or rotating the displayed information, demonstrating event-driven embedded control. Through this project, students gain hands-on experience in embedded vision systems, ARM-based edge-AI processing, FPGA-based multicore design, SPI communication, and hardware–software co-design using soft-core processors. To facilitate structured progress, the project is divided into three milestones. The marking scheme, rubrics, and deadlines for milestone demonstrations are provided in later sections.

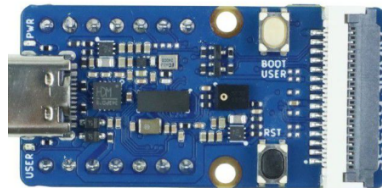


2026 - ECE3073 – Malaysia Mini Project

Coordinator, Lecturer: A/Prof. Narayanan Ramakrishnan

Lecturers : Dr. Howgen Kesuma and Dr. Chamalka Perara

2. Components

Item	Description
DE 10 Lite board with cable – 1 pc	
Groove AI Vision 2 (https://my.cytron.io/p-grove-ai-vision-module-v2?r=1)	
5MP OV5647 Camera Module for Raspberry Pi (https://my.cytron.io/p-5mp-camera-board-for-raspberry-pi)	
Crow-tail speaker (https://my.cytron.io/p-crowtail-speaker-2.0?r=1)	

LED Module

(<https://my.cytron.io/c-led-lcd/p-traffic-light-led-module>)

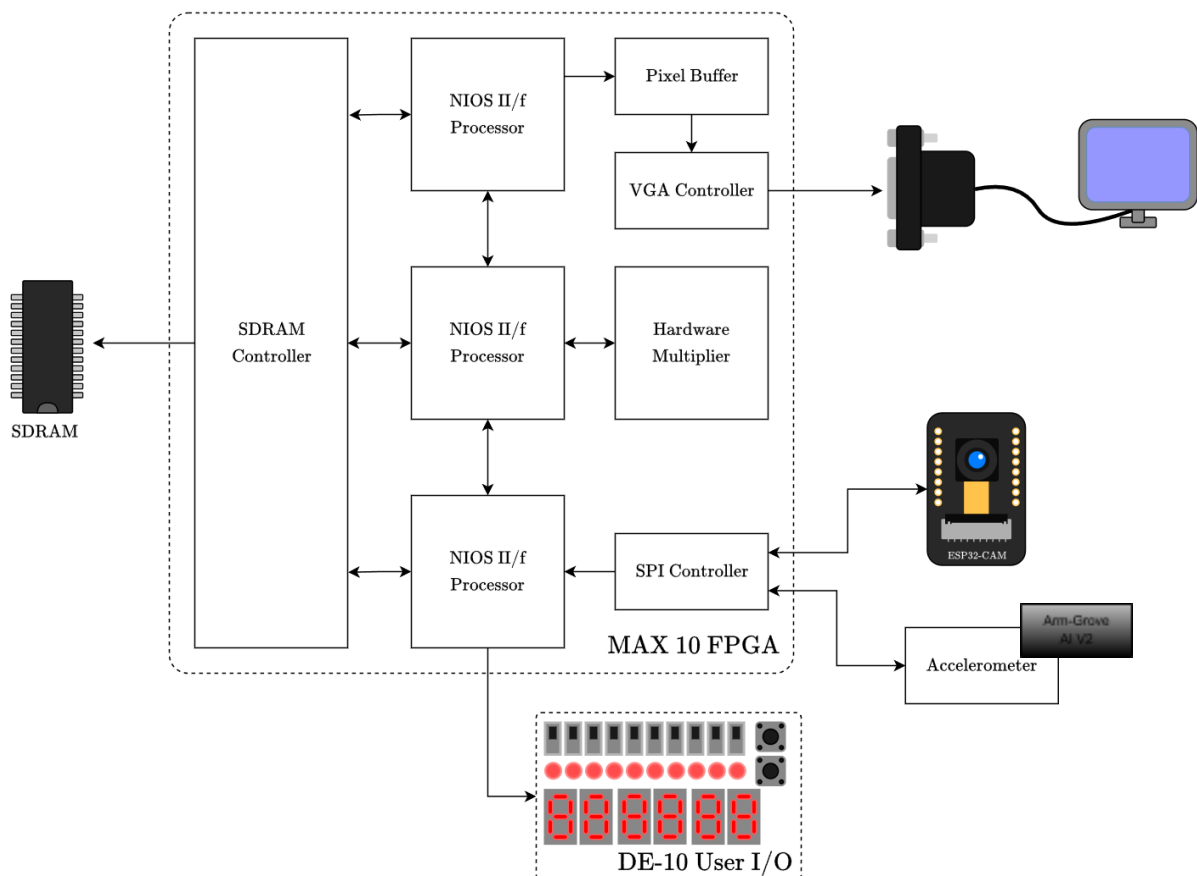


Figure 1: High-level overview of the final design

Figure 1 shows overall top level design required to be built: SDRAM, HEX Displays, LEDs and switches as required, VGA connectivity, Grove AI V2, Accelerometer to be interfaced with the NIOS system (one or more processor) through suitable controllers, NIOS PIO, NIOS UART, NIOS SPI.

2026 - ECE3073 – Malaysia Mini Project

Coordinator, Lecturer: A/Prof. Narayanan Ramakrishnan

Lecturers : Dr. Howgen Kesuma and Dr. Chamalka Perara

3. Task

The objective of this mini-project is to design and implement a real-time embedded vision and multitasking system using the Grove AI V2 vision module and the DE10-Lite FPGA platform. The system begins with the configuration and setup of the Grove AI V2 vision module and camera to capture alphanumeric text scrolling in front of the camera at varying speeds. The scrolling text contains meaningful information with embedded instructions that must be interpreted and executed by the NIOS II processors implemented on the FPGA.

The Grove AI V2 module performs on-device vision processing to convert the captured image or video stream into text information. The extracted text data is transmitted to the DE10-Lite FPGA system, where it is processed by multiple NIOS II soft-core processors operating in parallel. Based on the embedded instructions contained within the text, the system controls various hardware peripherals on the DE10-Lite board, including displays, LEDs, audio output, and external interfaces.

Each word in the scrolling text will consist of six letters or fewer, followed by a space. When Switch 2 is turned ON, the captured sentence must be displayed as a scrolling message on the display interface. While Switch 2 remains ON, the system must also display the NIOS II CPU utilization, expressed as a percentage, clearly indicating processor load and multitasking behaviour.

Hardware Control and Functional Requirements

The following hardware actions must be implemented as part of the system functionality:

Reference number	Hardware Component	Required Action
1	Switch 1	Controls the HEX display: OFF when Switch 1 is OFF, ON when Switch 1 is ON
	Switch 2	Displays the captured sentence as a scrolling message; also enables CPU load (%) display
	Switch 3	Displays CPU utilization (%)
	Switch 4	Turns the speaker ON or OFF
	Push Button 1	Captures the current image from the Grove AI V2 module and stores it in SDRAM
	Push Button 2	Displays the captured image on the VGA output along with the extracted text
	External Crowtail Speaker	Produces an audio tone corresponding to the frequency of the scrolling text
	LEDs	Operate according to instructions embedded in the captured text
	GPIO Pins 1 & 2	Optional outputs for oscilloscope connection to measure latency and timing
2	Grove AI V2 + Camera Module	Converts image/video input into text information
3	Accelerometer	Instruction to VGA for tilt display

2026 - ECE3073 – Malaysia Mini Project

Coordinator, Lecturer: A/Prof. Narayanan Ramakrishnan

Lecturers : Dr. Howgen Kesuma and Dr. Chamalka Perara

Reference number	Hardware Component	Required Action
	VGA	Interface to display

Embedded Instruction Execution

The scrolling text contains embedded control instructions that dictate hardware behavior. For example, the text may specify actions such as:

- Green LED blinking every 1 second
- Red LED blinking once every 2 seconds
- Enabling or disabling specific LEDs
- Modifying audio output characteristics

These instructions must be decoded and executed in real time by the NIOS II processors.

Multicore and Real-Time Processing Requirement

Students are required to implement more than one NIOS II soft-core processor within the FPGA to achieve true parallel processing. Tasks such as vision data handling, display management, LED control, audio generation, and system monitoring should be distributed across multiple processors.

At least one NIOS II core must run a Real-Time Operating System (RTOS) to manage timing-critical functions, such as periodic LED blinking, audio tone generation, and CPU load measurement. This requirement ensures that students demonstrate proper use of multitasking, scheduling, and real-time constraints within an FPGA-based embedded system.

Orientation-Based VGA Display Requirement

In addition to the above functionalities, the system must incorporate orientation-aware display control using the on-board accelerometer available on the DE10-Lite board. When the FPGA board is tilted or rotated, the accelerometer measurements must be used to dynamically adjust the orientation of the displayed image and text on the VGA output. The captured image and extracted text must be displayed in a defined orientation (e.g., portrait, landscape, rotated 90° or 180°) depending on the board's tilt angle. This feature demonstrates sensor-based event detection, real-time signal processing, and adaptive graphical output, and must be implemented as part of the multitasking embedded system architecture.

4. Phase, Milestone, and Marking Scheme

To aid with the scope of the project, the project has been separated into three phases and a gantt chart is provided as guidance to follow

Weeks

2026 - ECE3073 – Malaysia Mini Project

Coordinator, Lecturer: A/Prof. Narayanan Ramakrishnan

Lecturers : Dr. Howgen Kesuma and Dr. Chamalka Perara

Task and Phase	Milestone	3-4	5	6	7	8	9	10	11	12
Announcement and group discussion, Git Setup										
Phase -1: Lab 3 and 4, expanded for NIOS system setup and all hardware reference no. 1 interfaced.										
Phase -1: The interfaced hardware can handle instructions from NIOS(either by polling or interrupt)										
Push button 1 and 2 are programmed for a sentence manually available in SPI/UART										
Phase -1 and 2: single image stored to SD RAM, basic SPI and Accelerometer programming	Milestone (1): Demo (25 marks)				x1					
Phase - 2: Groove AI V2 programmed to capture a scrolling text via camera and converted to text.										
Phase 2 and 3: Atleast one more co-processor included and streamed data from Grove AI V2 is captured to one of the NIOS core. Hardware ref No. 2 and 3 also now interfaced with NIOS core.							x2			
Phase 3: RTOS and Multiprocessing improved, optimisation programming started, live image of Grove AI V2 now processed well .	Milestone (2) – 35 marks									
Phase 3: Groups optimising the multiprocessing and aim to achieve to capture highest possible text information at the shortest time.										
Phase 3: Students introducing their own innovation in creating applications beyond the task										

2026 - ECE3073 – Malaysia Mini Project

Coordinator, Lecturer: A/Prof. Narayanan Ramakrishnan

Lecturers : Dr. Howgen Kesuma and Dr. Chamalka Perara

requested						
Buffer week to complete pending , validation, checking before demo days.	Milestone (x)3 ; Final Demo (40 Marks)					x3

Phase - 1 : Milestone 1 (25 marks):

1. Milestone 1 focuses on system bring-up, hardware familiarisation, and initial NIOS II integration. Student groups must complete project announcement discussions, group formation, and Git repository setup. Using extended content from Lab 3 and Lab 4, students are required to implement the NIOS II system and integrate **all hardware components as specified in Figure 1**, ensuring that all defined process steps are implemented and demonstrable.
2. At this stage, all devices must be at least interfaced and operational using a polling-based control loop, although selected peripherals may be implemented using interrupt mechanisms at the group's discretion. Push Buttons 1 and 2 must be programmed to handle a manually defined sentence available through SPI or UART. A single image must be captured and stored in SDRAM, and basic SPI and accelerometer functionality must be demonstrated. VGA controllers, SPI, and UART must already be included in the Quartus design and compile without errors.
3. Direct Grove AI V2 image capture and inference are **not mandatory** in this milestone and will be completed in Phase 2. The ESP8266 module must be physically attached; however, Firebase integration is not required at this stage. During the demonstration, **all students in the group must be prepared to individually demonstrate** any part of the system, as assessors may request demonstrations from any group member.
4. GIT setup completed. Refer this [link](#)

Marking scheme and rubrics checklist for Milestone 1:

Component	Marks	Fail	Pass (50 % marks)	HD (Above 80 % or Full depending on interview)
GITHub setup	5	No Git setup as per instructions	-	Git Setup done
Quartus project compiles without errors and includes NIOS II system, VGA controller, SPI, UART, GPIO, SDRAM, and accelerometer interfaces as specified in	5	Most of the hardware not progressed beyond Lab 3 or 4	Errors showed up during the demo but Quartus has a design.	Successful download to FPGA no errors.

2026 - ECE3073 – Malaysia Mini Project

Coordinator, Lecturer: A/Prof. Narayanan Ramakrishnan

Lecturers : Dr. Howgen Kesuma and Dr. Chamalka Perara

Figure 1.				
All required hardware peripherals are interfaced and operational using polling-based control (interrupts optional), including switches, push buttons, LEDs, HEX display, and speaker	5	One or more required hardware peripherals are not interfaced or are non-functional.	All required hardware peripherals (switches, push buttons, LEDs, HEX display, and speaker) are correctly interfaced and operational using polling-based control, with optional use of interrupts.	All required hardware peripherals are robustly interfaced and operate reliably under polling-based control, with clean integration and consistent behaviour across demonstrations.
Push Buttons 1 and 2 correctly process a manually defined sentence via SPI or UART, and a single image is successfully captured and stored in SDRAM.	5	Polling/interrupt-based control is incomplete or unstable, with frequent manual intervention required.	Push Buttons 1 and 2 correctly process a manually defined sentence via SPI or UART, and a single image is successfully captured and stored in SDRAM, with minor delays or non-critical limitations.	Push Buttons 1 and 2 process the manually defined sentence accurately via SPI or UART, and image capture and SDRAM storage are performed efficiently and consistently, demonstrating stable timing and well-structured control logic.
All group members can independently explain and demonstrate any subsystem, including NIOS task flow, hardware interfacing, and basic system operation.	5	Independent Q&A – Most questions not answered/demonstrated	At least 50 % of questions answered well by all members	More than 80 % of questions answered by all members of the group.

2026 - ECE3073 – Malaysia Mini Project

Coordinator, Lecturer: A/Prof. Narayanan Ramakrishnan

Lecturers : Dr. Howgen Kesuma and Dr. Chamalka Perara

If a student is not available (without any prior permission) during the demo day, 0 will be awarded to the student, however the rest of the team members get marks.

Phase - 2 : Milestone 2: Vision Integration, Multicore Expansion, and Interrupts (35 Marks)

Milestone 2 builds on the foundational interfacing from Milestone 1 by integrating real-time vision processing, parallel task execution, and interrupt-driven control. By this stage, the project's Git repository must reflect significant updates, demonstrating consistent progress in coding, hardware integration, and testing.

The Grove AI V2 module now takes center stage. It must be programmed to capture scrolling text via its camera and convert it into digital text data, which is transmitted to the FPGA system using SPI or UART. A dedicated NIOS II processor is responsible for receiving and handling the Grove AI data stream. At this milestone, groups must demonstrate that captured text data can influence system outputs (such as controlling LEDs, displays, or audio).

Parallel processing is enhanced by adding at least one more NIOS II co-processor, allowing different tasks to run concurrently—e.g., one processor handles vision data while another manages hardware outputs. Groups should begin using interrupt mechanisms to improve system responsiveness, shifting some wafer process steps (originally in polling loops) to interrupt-driven routines. Lab 6 and the Week 8 workshop provide guidance for co-processor setup and interrupt handling.

Additionally, VGA output must be functional, displaying either captured text or system status information. The accelerometer interfacing must be completed via SPI, allowing the system to detect board tilt and implement an emergency stop. The stepper motor, if part of the setup, should be fine-tuned for precise control. Finally, latency in critical steps should be measured using GPIO pins and oscilloscope methods, as practiced in earlier labs.

By the end of Milestone 2, the system should show a robust combination of vision integration, multitasking with co-processors, and interrupt-driven process control.

M2 Marking Rubrics:

Criterion	Fail (<50%)	50% (Meets Minimum)	65% (Good)	80%+ (Very Good / Excellent)
1. Grove AI V2 Vision Integration (10 marks)	Grove AI V2 not integrated or data not reliably received by NIOS. Vision output	Grove AI V2 captures scrolling text and basic text data is received by	Vision-to-text conversion is reliable, text content is decoded correctly, and	Robust, low-latency vision-to-text integration with stable streaming and clear linkage to system actions.

2026 - ECE3073 – Malaysia Mini Project

Coordinator, Lecturer: A/Prof. Narayanan Ramakrishnan

Lecturers : Dr. Howgen Kesuma and Dr. Chamalka Perara

Criterion	Fail (<50%)	50% (Meets Minimum)	65% (Good)	80%+ (Very Good / Excellent)
	incomplete or incorrect.	NIOS via SPI/UART.	influences system behaviour.	
2. Multi-NIOS / Parallel Processing Architecture (10 marks)	Only a single NIOS core used or multicore design is non-functional.	At least two NIOS cores instantiated with basic task separation.	Clear task partitioning across multiple NIOS cores with parallel execution demonstrated.	Well-designed multicore architecture with effective load distribution and minimal contention.
3. Interrupts and Real-Time Behaviour(2 marks)	System relies entirely on polling with poor responsiveness.	Some interrupt usage introduced for selected tasks.	Interrupts are used meaningfully to improve timing and responsiveness of multiple steps.	Interrupt-driven design is well planned, improving determinism and real-time performance.
4. VGA, Accelerometer, and System Control(5 marks)	VGA or accelerometer not functional or incorrectly integrated.	VGA output works and accelerometer data is readable.	VGA displays text/image correctly and accelerometer enables system control (e.g., emergency stop or orientation).	Dynamic VGA behaviour with orientation-aware display and stable sensor-driven control.
Latency Measurement and Timing Analysis(3 marks)	No latency measurement or incorrect GPIO usage.	Latency measured for at least one process step using GPIO.	Multiple steps measured with reasonable explanation of timing impact.	Clear, repeatable latency measurements with analysis linked to system optimisation.
6. Git Usage and Project Management (5 marks)	Git not used properly or minimal commits with no structure.	Git repository updated with basic commits and code uploads.	Regular commits showing logical progress across subsystems.	Well-structured Git usage with meaningful commit messages, branches, and clear development history.

Week 9 Checkpoint :

To ease the progress, students should have at least completed 50 % of the above task to register the milestone progress, while they could still continue (carry forward) pending items to the M3 demo in week 12. However all members of the team should be present during checkpoint assessment in the lab to discuss with the lecturer on the progress. We strongly

2026 - ECE3073 – Malaysia Mini Project

Coordinator, Lecturer: A/Prof. Narayanan Ramakrishnan

Lecturers : Dr. Howgen Kesuma and Dr. Chamalka Perara

encourage students to finish all the tasks so that remaining weeks could be managed for whole project completion. Week 9 checkpoint marks gives you quantitative feedback of your progress.

Phase- 3: RTOS Integration, System Optimisation, and Final Demonstration (40 Marks)

Milestone 3 is the final phase of the mini-project and focuses on full system integration, real-time performance, and demonstration of advanced embedded system design. By this stage, the project Git repository must be well maintained and fully updated, clearly reflecting final integration, optimisation, and debugging activities.

A key requirement of this milestone is the accurate and efficient end-to-end processing of text captured by the Grove AI V2 module. The scrolling text detected by the Grove AI V2 must be correctly decoded and displayed across multiple output interfaces, including the HEX display, VGA monitor, LEDs, and speaker. The system should demonstrate meaningful and innovative use of these outputs, where the displayed text, LED behaviour, and audio output are logically linked and clearly derived from the captured text content. Groups that demonstrate higher scrolling speeds and higher frame-rate text capture, while maintaining correct and stable display on the HEX and VGA outputs, will be awarded higher marks.

Building upon the multiprocessing and interrupt-based designs developed in Milestones 1 and 2, this milestone requires the integration of a Real-Time Operating System (RTOS) on at least one NIOS II processor. The RTOS must be used to manage time-critical tasks such as text processing, display updates, audio generation, and sensor handling. Students should demonstrate how RTOS-based scheduling improves system responsiveness, timing predictability, and overall performance.

In addition, groups are expected to present student-led enhancements that extend beyond the minimum task requirements. These may include performance optimisation, enhanced visual presentation, improved user interaction, or novel system behaviour. The final demonstration must show a stable, high-performance embedded vision system, supported by clear explanations of system architecture, task partitioning, and optimisation strategies. The mini-project carries a total of 100 marks, contributing 25% to the overall unit assessment.

Criterion	Marks	Fail (<50%)	50% Level	65% Level	80%+ Level
1. Grove AI V2 Text Capture & High-Speed Scrolling Detection	15	Text capture unstable or incorrect; fails at moderate scrolling speeds; HEX/VGA	Text correctly captured and displayed at low scrolling speed on HEX or VGA.	Reliable text capture and display at moderate scrolling speeds on HEX and VGA	High scrolling speed text reliably captured , correctly decoded, and displayed in real time on HEX and

2026 - ECE3073 – Malaysia Mini Project

Coordinator, Lecturer: A/Prof. Narayanan Ramakrishnan

Lecturers : Dr. Howgen Kesuma and Dr. Chamalka Perara

Criterion	Marks	Fail (<50%)	50% Level	65% Level	80%+ Level
		output unreliable.		with minor errors.	VGA with stable operation.
2. Innovation & Multimodal Output Utilisation	10	No meaningful innovation beyond minimum requirements.	Basic use of LEDs or speaker linked to text.	Clear innovation using LEDs, speaker, and/or VGA to enhance application behaviour.	Strong, original innovation with well-synchronised HEX, VGA, LEDs, and audio creating a compelling application demo.
3. Git Usage & Project Management	5	Minimal or poorly organised Git usage; unclear progress history.	Git updated with basic commits and final code.	Regular commits reflecting logical development progress.	Well-structured Git usage with meaningful commit messages and clear evolution of the project.
4. RTOS Integration & Task Scheduling	5	RTOS missing or unstable.	RTOS integrated with basic task scheduling.	RTOS manages multiple tasks with priorities and stable execution.	RTOS used effectively to improve responsiveness, determinism, and system organisation.
5. System Stability & Final Demonstration Quality	5	System unstable; demo incomplete or unclear.	System functions under normal conditions with basic explanation.	Stable system with clear explanation of architecture and data flow.	Highly stable, polished final demo with confident technical explanation and stress-tested behaviour.

All required information and readings are provided with the individual document for each milestone.

Milestone 3 will end with a demonstration of your design along with an individual interview during your week 12 lab session as well as a 2-page report is **compulsory** during the demonstration. During the interview, all team members should be in the capacity to answer any question regarding the work they have submitted as a team.

Checklist for week 12

1. M3 activities completed and ready for demonstration
2. The 2 page report is ready for submission and discussion.

2026 - ECE3073 – Malaysia Mini Project

Coordinator, Lecturer: A/Prof. Narayanan Ramakrishnan

Lecturers : Dr. Howgen Kesuma and Dr. Chamalka Perara

3. After demonstration, marks given by lecturers are photographed and uploaded to moodle links.

Important Information

Estimated Time Commitment: 3-5 hours per week per student

Academic Integrity: The ECE3073 project is to be completed in **teams**. You can and should discuss the questions and concepts with other groups. However, **you may not:**

- write the code together with other groups,
- show other groups your HDL/Schematic,
- copy other groups' work, or
- present work you have found online as your own (i.e. without attribution),

This is plagiarism and/or collusion and is prohibited by Monash's academic integrity policy. Breaches of this policy attract serious consequences. Note that all code submitted to this unit will be run through the MOSS system to check for plagiarism.

If you research some concept online or re-use HDL from your labs (or the unit) you must cite it (any citation method is acceptable).

Generative AI:

As a reminder, you may use any AI tools at your disposal, in any capacity, to complete the assigned work. However, you must fully understand the work you submit for assessment. Any use of generative AI for this purpose must be appropriately acknowledged using the provided [AI disclosure form](#) and included in your milestone submission.

Due Dates

For submission instructions and the mark distributions, see each individual milestone's document.

Milestone 1:(25 marks)

Submissions and Demo Due: TBA

Milestone 2:(35 marks)

Submissions and Demo Due: TBA

Milestone 3:(40 marks)

Submissions Due and Demo Due: TBA

Late Submissions:

2026 - ECE3073 – Malaysia Mini Project

Coordinator, Lecturer: A/Prof. Narayanan Ramakrishnan

Lecturers : Dr. Howgen Kesuma and Dr. Chamalka Perera

Late submissions will be accepted until Wednesday / Thursday following the submission deadline without special consideration. The standard 5% penalty for every day that the assignment is late.

Special Consideration:

Special consideration requests are possible but only apply to a single student. Teams must submit milestone deliverables and complete the interview/demonstration by the specified deadlines. If a team member is granted a special consideration, they will receive one new time to demonstrate their work.

For Milestone 3 deliverables, a short-term extension of max 1 day is available to the whole team if it is not already used by all team members. Special consideration requests are possible but only apply to a single student. A 5% penalty per day after the specified deadline if no special consideration, but max by Thursday of following week, **if student not available during the demo day 0 will be awarded to the student, however rest of the team members get marks.**

Additional information on relation of milestones to learning outcomes

Milestone 1 aims to solidify your foundational knowledge in several key areas of computer systems:

- Managing memory for storing and reading data (LO2)
- Integrating software and hardware (LO1)
- Designing systems based on a high-level description and hardware constraints (LO2)
- Reading and handling hardware interrupts from within a processor (LO1)
- Manage repositories, track changes, collaborate using branches, and resolve merge conflicts in Git (LO4)

Milestone 2 aims to extend your understanding of the above areas as well as provide experience and further your skills in:

- Reading and handling hardware interrupts from within a processor (LO1) using version control software to manage large projects (LO4)
- Utilising industry-standard communications protocols (LO4)

Milestone 3 aims to solidify your understanding of the more advanced/challenging topics in computer systems

- Measuring and improving system performance through the use of multithreading, multiprocessing and hardware-based techniques (LO5)
- Documenting and providing relevant information to external stakeholders in a concise manner (LO2, LO4)
- Utilising version control software to manage large projects (LO4)